Fiscal Year:	FY 2005	**Task Last Updated:**	FY 07/22/2005
PI Name:	Archambeau, John Orin M.D.		
Project Title:	Quantification Of The Dose Response Of The Microvessel Parameters In Retina, Cortex And White Matter Following Iron-56 Irradiation And Proton Irradiation		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	92354	**Congressional District:**	41
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	02-OBPR-02
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No. of PhD Candidates:	0	**No. of Master' Degrees:**	2
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
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Key Personnel Changes/Previous PI:			
COI Name (Institution):	Mao, Xiao Wen (Radiation Biology Program Department of Radiation Medicine)		
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	A main concern of NASA managers is to define the risks of tissue damage of brain and eye for astronauts associated with exposure to galactic cosmic ray particle (GCR) irradiation during a proposed space mission to Mars. Research is needed to specify the tissue effects produced by the high Z, high energy (HZE) particles that are found in space and to define the dose response of these tissues. Current histopathologic methods used to evaluate tissue changes in late-responding, non-proliferative tissues such as the brain and vessel endothelium rely on observations, word descriptions and arbitrary scoring. The approach is considered to be qualitative and lacks rigor. The proposed research introduces a detailed, step-by-step stereology program into radiation biology that will quantify the dose response of tissue population changes in situ following irradiation. These methods are known to anatomists and materials scientists but have been ignored by radiation biologists until this laboratory used them to quantify the dose response of the non-proliferative retinal vasculature following single and split time-dose schedules of protons [1, 2]. The brain and retina		

Task Description:	microvessel endothelial cell population number, vessel length, and number of microvessel branching will be measured following single and split dose schedules of Iron-56 particles. The results will be compared with the changes produced by protons to define the dose response of the cortex and white matter microvessel endothelial populations and to specify the RBE quality factor of Iron-56 referenced to protons. Three-dimensional reconstructions of the microvessels of the cortex and white matter will be compared. Our objective is to utilize three dimensionsl reconstruction and stereology methods to quantify the microvessel population and vessel cell kinetic parameters in the cortex and white matter regions of the brain in an effort to explain the difference in dose response between the two regions. The significance of this research for NASA and for radiation biology is that it introduces into radiation biology a valuable quantitative technique that can be applied to quantify population morphologic, kinetic and identifiable functions in-situ following irradiation. It also defines the contribution (role) of the irradiated vasculature in producing radiation changes and the dose response of the brain and retina.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	These techniques used to quantify the dose response of the retina and the brain can be applied in future translational patient studies to determine the radiation dose response.
Task Progress:	In the use of confocal laser scanning the optical stacks are precisely oriented and aligned with fixed dimensions. In forming a mosaic image of the surface of the coronal brain slice, adjacent image stacks are linked, forming a confluent, red and green image of the brain slice showing the anatomic arrangement and the network of microvessels, larger vessels and anatomic variations The resulting mosaic images reveal the relationship of the microvessels across the coronal sections. The vessels in the cortex are branched with a hexagonal-like topology. The microvessels in the white matter are linear with few branches. These configurations are confirmed on the 3-d reconstructions. The stereology methods were used to estimate the volume density (#/mm3) of the parenchymal nuclei, endothelial cells and pericyte nuclei. These methods were also used to estimate the microvessel volume length density (mm length /mm3) in unirradiated brains. The average cell number density of the parenchymal nuclei (neurons, microglia, astrocytes and oligodendrocytes) was 588 nuclei/ mm3 in the white matter (resulting from the large number of oligodendrocyte nuclei packed into the myelin tracks. The toal number was 283 nuclei/mm3 in the cortex. There were an estimated 26 endothelial cell/mm3 in the white matter and 17 endothelial cell nuclei/mm3 in the cortex, essentially fewer nuclei per unit length. The value for pericytes was estimated to be 47/mm3 in the white matter and 58 /mm3 in the cortex. The length density of the microvessels was estimated to be 699 mm/mm3 (1/mm^2) in the white matter and about twice that, 1283 mm/mm3 (1'mm^2) in the cortex
Bibliography Type:	Description: (Last Updated: 10/06/2006)
Papers from Meeting Proceedings	Archambeau JO, Mao XW, Kubinova L, Archambeau P, Baqai W, Tormori Z. "Quantification of the dose response of the microvessel parameters in the rat retina, cortex and white matter following Iron-56 and proton irradiation." Bioastronautics Investigators Workshop, Galveston, Texas, January 2005. Bioastronautics Investigators Workshop, Galveston, Texas, January 2005. , Jan-2005
Papers from Meeting Proceedings	Archambeau JO, Mao XW, Kubinova L, Archambeau P, Baqai W, Tormori Z. "Rat Brain Microvessel 3-D Configuration in the Non-irradiated Cortex and White Matter: An Explanation for a Previously Observed Histology." 16th Annual NASA Space Radiation Investigatiors' Workshop, May 2005. 16th Annual NASA Space Radiation Investigatiors' Workshop, May 2005. , May-2005